

7-1 Write Equations

Equation Detective Agency | Standard: 6.EE.6 | Color-by-Number Practice

Learning Goal: I can write an equation to represent a real-world situation.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: variable, equation, equal sign

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Write an equation: A number plus 7 is 15. Work: _____ Answer: _____	2. Write an equation: A number minus 5 is 12. Work: _____ Answer: _____
3. Write an equation: 3 times a number is 24. Work: _____ Answer: _____	4. Write an equation: A number divided by 4 is 9. Work: _____ Answer: _____
5. Write an equation: 8 more than a number is 20. Work: _____ Answer: _____	6. Write an equation: 6 less than a number is 11. Work: _____ Answer: _____
7. Write an equation: 5 times a number is 45. Work: _____ Answer: _____	8. Write an equation: A number divided by 6 is 7. Work: _____ Answer: _____
9. Write an equation: Twice a number is 18. Work: _____ Answer: _____	10. Write an equation: A number plus 13 is 30. Work: _____ Answer: _____
11. Write an equation: A number minus 9 is 16. Work: _____ Answer: _____	12. Write an equation: 10 times a number is 90. Work: _____ Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

7-1 Mystery Picture: Detective Badge Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
$x + 7 = 15 = \text{red}$	$x - 5 = 12 = \text{orange}$	$3x = 24 = \text{yellow}$	$x/4 = 9 = \text{green}$
$x + 8 = 20 = \text{blue}$	$x - 6 = 11 = \text{purple}$	$5x = 45 = \text{pink}$	$x/6 = 7 = \text{brown}$
$2x = 18 = \text{gray}$	$x + 13 = 30 = \text{teal}$	$x - 9 = 16 = \text{black}$	$10x = 90 = \text{light blue}$

2	2	1	2	2	2	3	3	11	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
1	5	8	9	8	8	5	9	9	12	9	6
3	6	10	2	2	1	2	2	4	2	11	6
3	6	9	2	5	5	5	7	4	2	9	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	11	3	6	12	1	10	4	2	3	8
5	8	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	5	5	3	11	8
7	10	3	4	4	2	3	3	4	5	2	10
7	12	12	1	11	11	12	12	12	2	2	11
9	9	9	8	6	7	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	$x + 7 = 15$	red	Use a variable for the unknown.
2	$x - 5 = 12$	orange	Use a variable for the unknown.
3	$3x = 24$	yellow	Use a variable for the unknown.
4	$x/4 = 9$	green	Use a variable for the unknown.
5	$x + 8 = 20$	blue	Use a variable for the unknown.
6	$x - 6 = 11$	purple	Use a variable for the unknown.
7	$5x = 45$	pink	Use a variable for the unknown.
8	$x/6 = 7$	brown	Use a variable for the unknown.
9	$2x = 18$	gray	Use a variable for the unknown.
10	$x + 13 = 30$	teal	Use a variable for the unknown.
11	$x - 9 = 16$	black	Use a variable for the unknown.
12	$10x = 90$	light blue	Use a variable for the unknown.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.